SYSTEM ARCHITECTURE & METHODOLOGY DESIGN

Course: 23AD551 - Computer Networking Laboratory | NetLab-Pro Dynamic Campus Subnetting Suite

1. USER CONFIGURATION & INPUT MODULE

USER INPUT

User-Driven Requirements & Network Constraints

- Base IPv4 Network Scope (e.g., 10.50.0.0/16, 192.168.0.0/24)
- Dynamic Lab Quantity (1 to 32 Laboratories Supported)
- Lab Identity: Name, Department Wing, Workstation PC Count
- Future Host Growth Buffer (0% to 100% capacity reservation)
- Interactive Supernet Selection (Contiguous Member Labs to Merge)
- Diagnostics Input: Endpoint Pair IPs for L2/L3 Path Verification

User Inputs

2. WEB PRESENTATION & APPLICATION SERVER

FRONT-END

Python Flask Controller & Dynamic Modern Web UI

- Single-Page Dashboard: Responsive Tailwind CSS & UI Panels
- Interactive Setup Wizard: Multi-Lab Real-time Parameter Sizing
- REST API Controllers: /api/state, /api/supernet, /api/binary
- Real-time DOM Data-Binding & Dynamic Capacity Utilization Bar
- Interactive Supernetting: 1-Click Lab Merge & 1-Click Split/Revert
- Client-side Input Validation & Async Fetch API Communications

High-Speed C Foreign Function Calls

3. CROSS-PLATFORM FFI INTEROPERABILITY BRIDGE (ctypes)

Direct Memory Binding Between Python Web Layer & Native Compiled C Dynamic Library

Memory Marshaling	JSON Buffer Exchange	Platform Detection
C Structs <-> Python Objects	Fast In-Memory Serialization	Loads .dll (Win) / .so (Linux)

Native Binary Execution

4. C NATIVE COMPUTATIONAL KERNEL (bin/netlab_core.dll | netlab_core.so)

C BACKEND

High-Performance Bitwise Algorithms, Descending Bin-Packing & Routing Logic in Pure C (ANSI C99)

A. VLSM Engine

Descending Bin-Packing

- qsort() Descending Host Sort
- Host Bit Math: (2^h - 2) >= Req
- Power-of-2 Block Alignment
- Prefix Sizing: CIDR = 32 - h
- Subnet Boundary Clamping
- Address Space Exhaustion Guard

B. Supernetting

LCP Route Summarization

- Contiguity Verification (0 Gaps)
- XOR LCP: (min_net ^ max_bcast)
- Supernet Mask & Prefix Math
- Route Reduction % Efficiency
- Power-of-2 Block Condition
- Cisco 'ip route' Command Gen

C. Binary Boolean

32-Bit Bitwise Algebra

- Bitwise AND: Host & Subnet Mask
- Network ID & Broadcast Derivation
- Usable Host Boundary Slicing
- 32-Bit Octet Dot Separation
- Borrowed Bits Classification
- RFC 1918 Scope Identification

D. L2/L3 Diagnostics

Reachability Analyzer

- Local Subnet Check (Direct ARP)
- Inter-VLAN L3 Router Gateway
- Asymmetric Mask Blackhole Check
- Reserved Host IP Validation
- Wire-Speed vs Hop Diagnosis
- VLAN SVI Remediation Advice

Deterministic Subnet Allocations & Route Summaries

5. APPLICATION DELIVERABLES & 4-TIER CAMPUS TOPOLOGY MAPPING

OUTPUTS

Essential Deliverables Generated for Enterprise Deployment & Lab Report

Subnet Matrix Table

- Network & Broadcast IPs
- First - Last Usable Hosts
- VLAN IDs & SVI Gateways
- Allocation Efficiency %

Workstation Fleet

- PC-by-PC Hostnames & IPs
- Assigned MAC Addresses
- DHCP Start / End Pools
- Lease Capacity Counters

Cisco Config (.cfg)

- VLAN 802.1Q Database
- Interface SVI IP / Masks
- DHCP Pool Configurations
- Static Route Summaries

Campus 4-Tier Topology

- Tier 1: Border Edge Router
- Tier 2: Core L3 Switch (SVI)
- Tier 3: Distribution Trunks
- Tier 4: Lab Access VLANs

Figure 2.1: System Architecture and Detailed Design Workflow of NetLab-Pro Campus Subnet Suite